Technische Hochschule Ingolstadt

Fakultät Maschinenbau

Autonomes Fliegen

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Prof. Dr. Gerhard Elsbacher,

and Dr. techn. Babak Salamat

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Kapitel

- **Kapitel 1: Übersicht über Unmanned Aerial Systems (UAVs) (1 W)**
- **Kapitel 2: Wesentliche Komponenten/Subsysteme eines UAV (1 W)**
- **Kapitel 3: Modellierung und Simulation von UAVs (1 W)**
- **Kapitel 4: Flight State Controller (PD Cascade) (1W)**
- **Kapitel 5: Pfadplanung (4 W)**
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- **Kapitel 8: Multi-Vehicle Cooperation and Coordination (1 W)**

Summary

Trajectory Planning

- **Introduction**
- **Polynomial trajectories**
- **A minimum-jerk trajectory**
- **Optimization**

Minimum-Jerk Trajectories

Neville Hogan (1984a) noted that smoothness can be quantified as a function of jerk, which is the time derivative of acceleration. Hence, jerk is the third time derivative of location (*i.e.*, position). If the location of a system is specified by variable $x(t)$, then the jerk of that system is:

$$\text{jerk} \quad \dddot{x}(t) = \frac{d^3 x(t)}{dt^3}$$

For your CNS to move your hand or some other end effector smoothly from one point to another, it should minimize the sum of the squared jerk along its trajectory. For a particularly trajectory $x_1(t)$ that starts at time t_i and ends at time t_f, you can measure smoothness by calculating a jerk cost:

$$\int_{t=t_i}^{t_f} \dddot{x}_1(t)^2 \, dt$$

Problem Formulation

To make the issue concrete, imagine that you wish to move something 10 cm in a 0.5 s period. The object will be at rest at start time and at the end of the movement. You can write this as:

$$\text{Properties of } x(t): \begin{cases} x(0)=0 & x(0.5)=10 \\ \dot{x}(0)=0 & \dot{x}(0.5)=0 \\ \ddot{x}(0)=0 & \ddot{x}(0.5)=0 \end{cases}$$

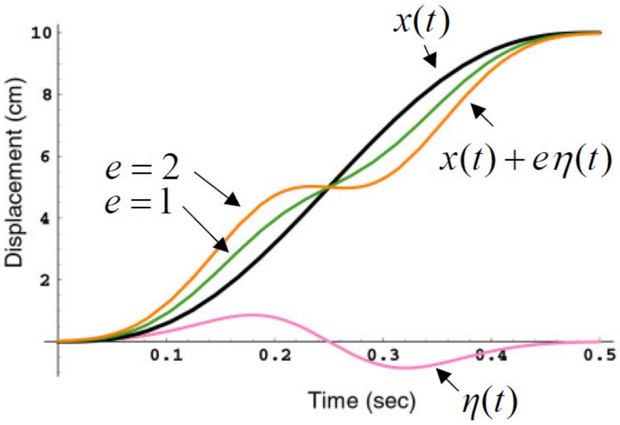
What trajectory $x(t)$ has the smoothest path? To find $x(t)$, you need to assign a jerk cost to each possible trajectory, and then find the trajectory with the least cost. Mathematically, this calculation corresponds to minimizing the **functional**:

$$H(x(t)) = \frac{1}{2}\int_{t=0}^{0.5} \dddot{x}^2 dt$$

Jerk Cost Functional (1D)

- The idea resembles finding the minimum of a function: you find the derivative of the function with respect to a small perturbation and when that derivative is zero, you have found a minimum.

$$\text{Properties of } \eta(t): \begin{cases} \eta(0)=0 & \eta(0.5)=0 \\ \dot{\eta}(0)=0 & \dot{\eta}(0.5)=0 \\ \ddot{\eta}(0)=0 & \ddot{\eta}(0.5)=0 \end{cases}$$



Closed-Form Minimum-Jerk (1D) 1

To minimize $H(x(t))$, you can replace $x(t)$ by a variation $x(t) \mapsto x(t) + e\eta(t)$ and you can find the derivative of the new functional with respect to the variation.

$$H(x(t)) = \frac{1}{2}\int_0^{0.5} \dddot{x}(t)^2 dt$$

$$x(t) \mapsto x(t) + e\eta(t)$$

$$H(x+e\eta) = \frac{1}{2}\int_0^{0.5} (\dddot{x} + e\dddot{\eta})^2 dt$$

$$\frac{dH(x+e\eta)}{e} = \int_0^{0.5} (\dddot{x} + e\dddot{\eta})\dddot{\eta}\ dt$$

$$\left.\frac{dH(x+e\eta)}{e}\right|_{e=0} = \int_0^{0.5} \dddot{x}\dddot{\eta}\ dt$$

Using integration by parts, you can rewrite this integral as:

$$\int_0^{0.5} \dddot{x}\dddot{\eta}\ dt = \int_0^{0.5} u\ dv = uv\Big|_0^{0.5} - \int_0^{0.5} v\ du$$

$$u = \dddot{x},\ \ dv = \dddot{\eta}\ dt,\ \ du = x^{(4)}dt,\ \ v = \ddot{\eta}$$

$$\int_0^{0.5} \dddot{x}\dddot{\eta}\ dt = \dddot{x}\ddot{\eta}\Big|_0^{0.5} - \int_0^{0.5} \ddot{\eta}\ x^{(4)}dt = -\int_0^{0.5} \ddot{\eta}\ x^{(4)}dt$$

where $x^{(4)}$ means 4th derivative of function $x(t)$. Continuing the integration by parts,

Closed-Form Minimum-Jerk (1D) 2

$$-\int_0^{0.5} \ddot{\eta}\, x^{(4)} dt = -\int_0^{0.5} u\ dv = -uv\Big|_0^{0.5} + \int_0^{0.5} v\ du$$

$$u = x^{(4)},\quad dv = \ddot{\eta}\ dt,\quad du = x^{(5)} dt,\quad v = \dot{\eta}$$

$$-\int_0^{0.5} \ddot{\eta}\, x^{(4)} dt = -x^{(4)}\dot{\eta}\Big|_0^{0.5} + \int_0^{0.5} \dot{\eta}\, x^{(5)} dt = \int_0^{0.5} \dot{\eta}\, x^{(5)} dt$$

$$\int_0^{0.5} \dot{\eta}\, x^{(5)} dt = x^{(5)}\eta\Big|_0^{0.5} - \int_0^{0.5} \eta\, x^{(6)} dt = -\int_0^{0.5} \eta\, x^{(6)} dt$$

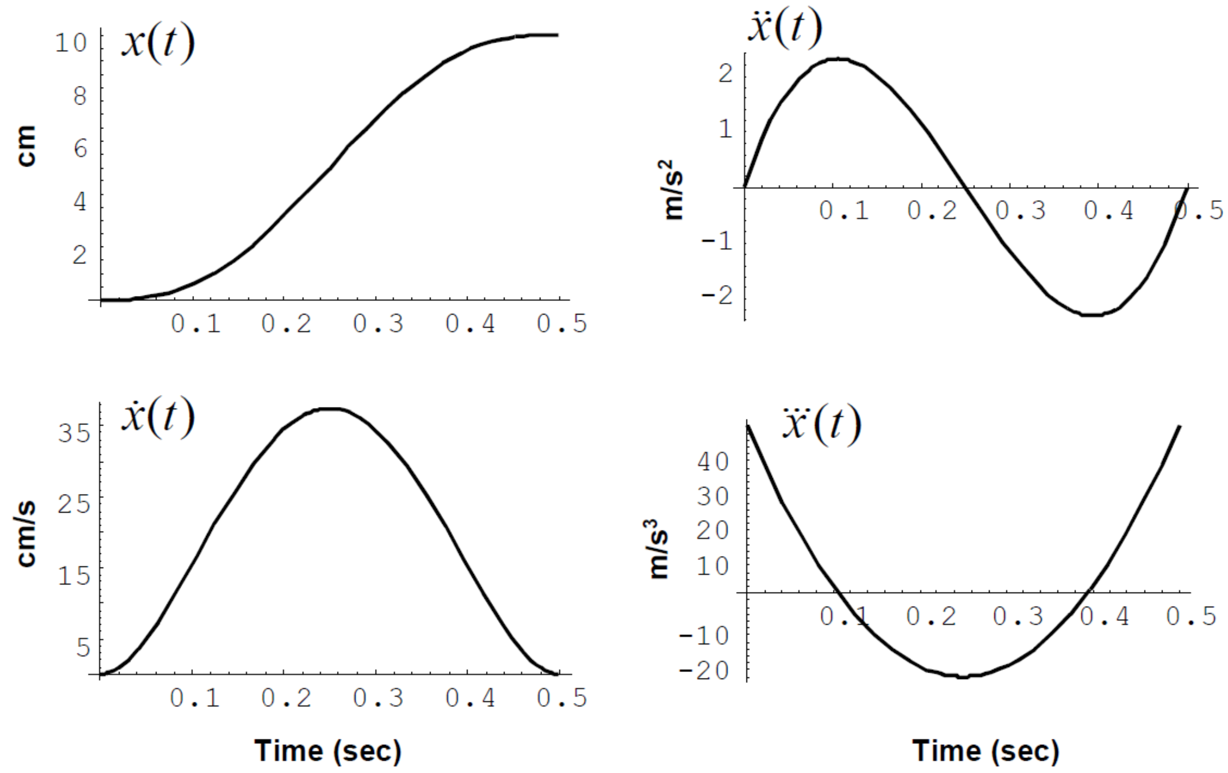
This final integral is the derivative of your functional, and you have:

$$\left.\frac{dH(x+e\eta)}{e}\right|_{e=0} = -\int_0^{0.5} \eta x^{(6)} dt \equiv 0$$

The above property must hold true for any function $\eta(t)$, and therefore you have the fact that

$$x^{(6)} = 0\,,$$

Closed-Form Minimum-Jerk (1D) 3



Multi-Dimensional Extension

Flash and Hogan (1985) found that, for end-effector locations specified as a vector of two or more dimensions, Eq. (1) described the minimum jerk trajectory for each dimension. For example, for a movement in two dimensions, the functional to minimize is:

$$H(\underline{x}(t)) = \frac{1}{2}\int_{t=0}^{a} (\dddot{x}^2 + \dddot{y}^2)\,dt$$

and the minimum jerk trajectory in two dimensions is:

$$\underline{x}(t) = \begin{bmatrix} x_i + (x_f - x_i)\left(10(t/a)^3 - 15(t/a)^4 + 6(t/a)^5\right) \\ y_i + (y_f - y_i)\left(10(t/a)^3 - 15(t/a)^4 + 6(t/a)^5\right) \end{bmatrix}$$

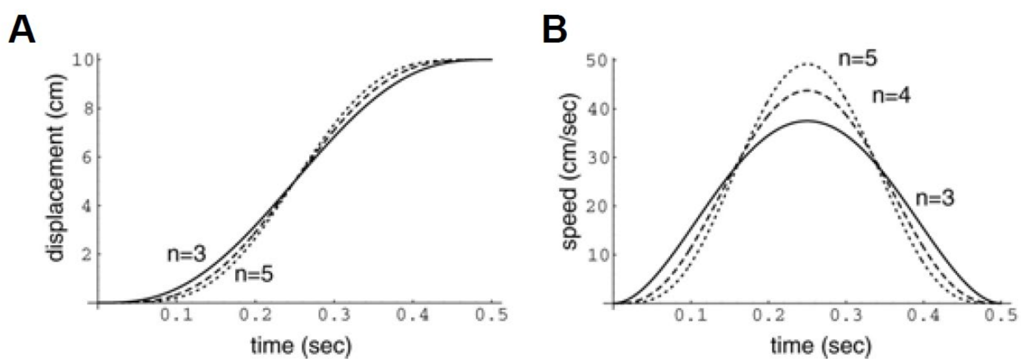
Why not minimum snap, crackle, or pop?

The third derivative of location with respect to time is called jerk. The fourth, fifth, and sixth derivatives are called snap, crackle, and pop, respectively. How can you know that a minimum-jerk description provides the best description of your reaching movements: Why not minimum snap?

To answer this question, Magnus Richardson and Tama Flash (2002) considered how $x(t)$ changed as a function of n in the following expression:

$$H(x(t)) = \frac{1}{2}\int_{t_i}^{t_f}\left(\frac{d^n x}{dt^n}\right)^2 dt$$

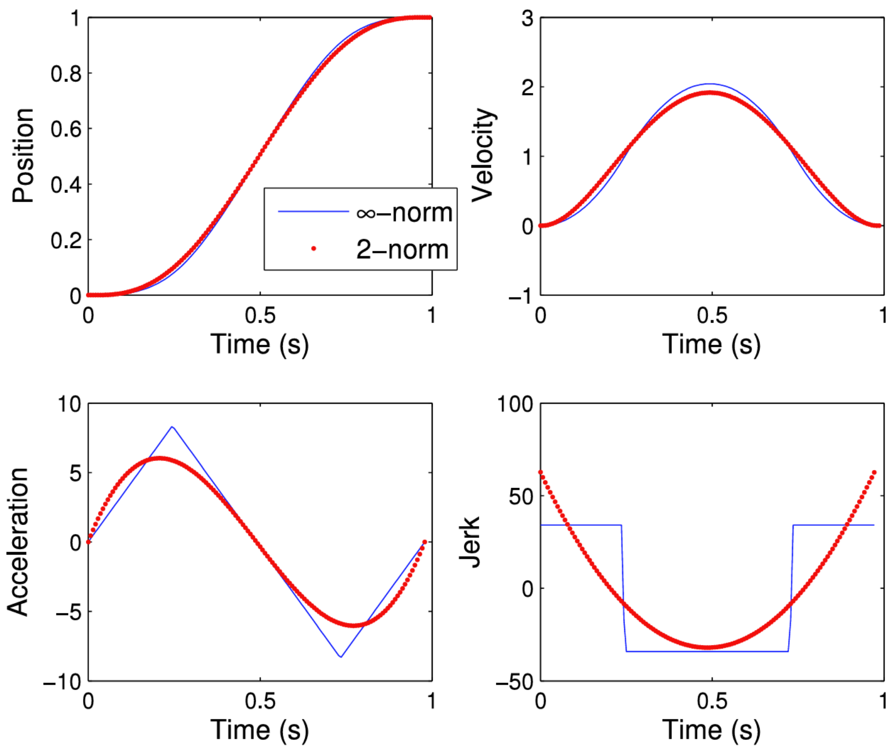
They found that as the order of the derivative n increased, the solution to the functional $x(t)$ approached a step function.



Feedback Controller View

Recall that the general solution to the minimum jerk functional was of the form $x^{(6)} = 0$, which implies that the trajectory is a fifth order polynomial. Let us normalize our measure of time so that it is 0 when we start the movement and 1 when we reach the target. If τ represents this normalized time, we have:

$$\tau = \frac{t - t_0}{D} \qquad D = t_f - t_0$$



State Space Form 1

Our minimum jerk trajectory has the form:

$$x(t) = a_0 + a_1\tau + a_2\tau^2 + a_3\tau^3 + a_4\tau^4 + a_5\tau^5$$

Note that $\dfrac{d\tau}{dt} = \dfrac{1}{D}$ and therefore:

$$\dot{x}(t) = \frac{a_1}{D} + \frac{2a_2}{D}\tau + \frac{3a_3}{D}\tau^2 + \frac{4a_4}{D}\tau^3 + \frac{5a_5}{D}\tau^4$$

$$\ddot{x}(t) = \frac{2a_2}{D^2} + \frac{6a_3}{D^2}\tau + \frac{12a_4}{D^2}\tau^2 + \frac{20a_5}{D^2}\tau^3$$

Rather than assuming that the movement begins from rest, we assume that our initial conditions are more generally specified as follows:

$$x(t_0) = x_i \quad \dot{x}(t_0) = v_i \quad \ddot{x}(t_0) = p_i$$

At $t = t_0$, we have $\tau = 0$. Therefore, we have:

$$a_0 = x_i \quad a_1 = Dv_i \quad a_2 = \frac{Dp_i}{2}$$

Our conditions at the end of the movement are:

$$x(t_f) = x_f \quad \dot{x}(t_f) = 0 \quad \ddot{x}(t_f) = 0$$

At $t = t_f$, we have $\tau = 1$. Therefore, we have:

$$a_3 = \frac{-3D^2}{2}p_i - 6Dv_i + 10(x_f - x_i)$$

$$a_4 = \frac{3D^2}{2}p_i + 8Dv_i - 15(x_f - x_i)$$

$$a_5 = -\frac{D^2}{2}p_i - 3Dv_i + 6(x_f - x_i)$$

State Space Form 2

This gives us an expression for $x(t)$ that is valid for any initial condition. For example, at any time into the movement t, label that time $t = t_0$, and assume that we are at state $q = \begin{bmatrix} x_i & v_i & p_i \end{bmatrix}^T$. The **change** that should occur in our acceleration is specified by $\dddot{x}$:

$$\dddot{x}(t) = \frac{6a_3}{D^3} + \frac{24a_4}{D^3}\tau + \frac{60a_5}{D^3}\tau^2$$

At $t = t_0$, $\tau = 0$, and $D = t_f - t$. Therefore

$\dddot{x}(t_0) = \frac{6a_3}{D^3} = \frac{60}{D^3}(x_f - x(t_0)) - \frac{36}{D^2}\dot{x}(t_0) - \frac{9}{D}\ddot{x}(t_0)$. If we write a control law as:

$\dot{q} = Aq + Bx_f$, then we have:

$$\dot{q} = \begin{bmatrix} \dot{x} \\ \ddot{x} \\ \dddot{x} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ \frac{-60}{D^3} & -\frac{36}{D^2} & -\frac{9}{D} \end{bmatrix} \begin{bmatrix} x \\ \dot{x} \\ \ddot{x} \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{60}{D^3} \end{bmatrix} x_f$$